

CROWSNEST, AB, Canada

WMO: 712360

Lat: **49.630N**Lon: **114.480W**Elev: **1303**StdP: **86.62**Time Zone: **-7.00 (NAM)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.5	-25.2	-33.6	0.2	-29.1	-29.2	0.3	-24.7	12.3	-2.0	11.4	-2.1	1.3	130	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	28.7	15.2	26.9	14.7	25.1	14.2	17.1	24.9	16.1	24.0	15.1	23.0	4.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.5	12.1	19.4	13.2	11.1	18.4	12.0	10.3	17.4	53.4	24.9	49.9	24.0	46.9	23.2	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.0	8.0	-35.1	31.8	3.9	1.7	-37.8	33.0	-40.1	34.0	-42.3	35.0	-45.1	36.2		
(5)			-35.2	19.1	3.8	1.0	-37.9	19.9	-40.2	20.5	-42.3	21.1	-45.1	21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-5.8	-5.6	-1.6	3.0	7.7	11.4	15.3	14.3	9.9	4.2	-1.4	-6.2
	DBStd	9.37	8.21	7.42	6.60	4.33	3.74	2.95	2.79	2.78	3.91	4.79	6.74	7.51
	HDD10.0	2689	491	436	359	212	92	18	2	3	48	186	342	502
	HDD18.3	5302	749	669	618	461	331	207	98	128	252	439	592	761
	CDD10.0	436	0	0	0	1	20	61	166	136	46	5	1	0
	CDD18.3	7	0	0	0	0	0	0	5	2	0	0	0	0
	CDH23.3	799	0	0	0	1	18	57	362	287	72	2	0	0
	CDH26.7	174	0	0	0	0	2	11	88	62	11	0	0	0
Wind	WSAvg	3.2	4.2	3.6	3.5	3.1	2.8	2.8	2.8	2.4	2.6	3.3	3.8	3.9
Precipitation	PrecAvg	696	51	42	50	51	79	97	49	46	51	51	63	53
	PrecMax	924	79	72	98	95	153	258	167	134	153	89	139	116
	PrecMin	510	16	12	17	29	14	38	8	9	6	20	16	24
	PrecStd	126	17	18	21	16	37	50	39	28	33	24	35	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	9.5	14.4	20.5	25.6	28.4	31.4	30.8	28.3	21.1	14.3	8.1
		MCWB	4.9	3.4	5.8	9.5	12.2	15.9	16.7	15.6	14.0	11.5	7.4	4.5
	2%	DB	6.6	6.9	11.0	17.5	22.8	24.7	29.1	28.4	25.4	17.4	11.1	5.9
		MCWB	2.7	2.5	4.7	8.0	11.3	14.2	15.7	14.8	13.4	9.5	6.2	2.7
	5%	DB	5.0	5.3	9.0	14.2	20.1	22.6	27.2	26.6	23.0	14.8	8.9	4.4
		MCWB	1.6	1.6	3.8	6.2	10.4	12.8	15.4	14.4	12.6	8.1	5.0	1.6
	10%	DB	3.8	3.7	6.9	11.4	17.5	20.2	25.2	24.7	20.0	12.4	6.9	2.9
		MCWB	0.9	0.4	2.7	5.1	9.3	11.8	14.7	14.0	11.4	7.0	3.5	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	4.4	7.1	10.1	13.6	17.3	19.1	17.8	15.8	12.3	8.8	5.2
		MCDB	8.9	7.5	12.2	18.7	22.5	25.0	26.7	25.7	24.3	18.2	12.2	7.6
	2%	WB	3.2	3.1	5.4	8.3	12.3	15.3	17.7	16.5	14.1	10.2	6.8	3.2
		MCDB	5.8	6.3	9.9	16.1	20.6	23.1	25.3	24.6	22.8	16.1	10.7	5.3
	5%	WB	2.1	1.9	4.2	6.9	11.1	13.9	16.6	15.6	13.0	8.7	5.3	1.7
		MCDB	4.5	5.0	8.2	13.2	18.5	20.4	24.2	23.8	21.7	13.9	8.5	4.0
	10%	WB	1.0	0.6	3.0	5.5	9.8	12.6	15.6	14.7	11.8	7.2	3.8	0.4
		MCDB	3.5	3.5	6.6	10.8	16.3	18.5	23.2	22.9	19.4	12.0	6.6	2.8
Mean Daily Temperature Range	5% DB	MDBR	10.0	11.3	11.2	12.3	13.8	13.6	16.6	17.4	15.3	11.9	9.1	9.2
		MCDBR	9.0	11.1	13.6	19.1	21.4	20.2	21.8	23.1	23.2	17.2	9.9	7.5
		MCWBR	6.5	7.6	8.1	10.3	10.7	10.1	9.9	10.7	11.6	9.5	6.3	5.6
	5% WB	MCDBR	8.0	10.8	12.6	17.3	18.9	17.6	18.8	19.8	21.2	16.0	9.5	7.2
		MCWBR	6.1	7.7	8.1	9.7	9.7	9.4	9.5	9.9	11.0	9.3	6.4	5.6
Clear-Sky Solar Irradiance	taub	0.239	0.250	0.278	0.331	0.339	0.331	0.330	0.341	0.308	0.281	0.259	0.231	
	taud	2.386	2.363	2.376	2.289	2.395	2.462	2.477	2.444	2.517	2.574	2.504	2.415	
	Ebn at Noon	841	915	941	914	919	927	922	895	898	873	814	805	
	Edh at Noon	62	82	97	119	113	106	103	102	85	67	56	52	
All-Sky Solar Radiation	RadAvg	1.10	2.07	3.12	4.49	5.33	5.75	6.72	5.52	3.84	2.30	1.16	0.86	
	RadStd	0.10	0.17	0.28	0.36	0.45	0.47	0.53	0.40	0.41	0.25	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page</